

The Games Issue — Artometrics

The Games Issue

Steam, Pokémon, boards, and fandom as preference infrastructure.

Which Steam Games Earned the Most Playtime Per Dollar?

Steam Spy-era metadata turns the storefront into a scored catalog. This file holds 26,688 games spanning 2004–2018, with a median Metascore of 73.0 and a high of 98.0. DEEP SPACE WAIFU: NEKOMIMI appears in the fact-box extremes; the 0 .. 20,000 owners band is the most common ownership bucket.

The question is how critic Metascore sits across time, titles, and owner tiers — and whether score and price move together. The calibration point is 73.0: above it, the critically strong; below it, the long shelf of ordinary releases.

Fast facts

The numbers that set the scale for this report:

26,688Records in the working dataset

73.0Median Metascore

98.0Highest observed Metascore

DEEP SPACE WAIFU: NEKOMIMITop Game by Metascore

2004–2018Year span covered in the file

0 .. 20,000Most common Owners

Data and method

The source is the TidyTuesday release from 2019-07-30 (R for Data Science community). The working file contains 26,688 rows and 11 columns after merging available tables in the week folder. Metascore is the primary metric; owners is the categorical size axis; price appears in the scatter.

Medians are used because scores and prices both skew. Index-style fields are excluded from metric selection.

Median Metascore edged up across the store's expansion years

Median Metascore Over Time

Median Metascore rises from 70.0 in the opening period to 73.0 at the close — landing on the file median. The typical scored game in this extract ends the window a few points higher than it began.

A rising median can mean stronger games, changing coverage of what receives scores, or both. The chart records the center's drift.

Little Triangle tops a tightly packed critical elite

Little Triangle leads at 98.0 — 95.5 marks the median among the top dozen

Little Triangle leads at 98.0, while 95.5 marks the median among the top dozen. The elite band is compressed near the ceiling of 98.0 — high scores clustered together rather than a lonely outlier.

Fact boxes also surface DEEP SPACE WAIFU: NEKOMIMI among notable titles in the ranking rules used there. The leaders chart makes the numeric top of this Metascore cut explicit.

Owner tiers carry different Metascore distributions

Metascore by Owners

Category boxes by owners show whether Metascore consensus is shared or contested across popularity bands. Games with millions of owners are not the same statistical population as the 0 .. 20,000 band that dominates row counts.

Volume of titles in the smallest owner bucket is a catalog fact; the boxes ask how scores spread inside each ownership tier.

Mid-million owner games clear the median; tiny audiences trail

Metascore vs median by Owners

The 2,000,000 .. 5,000,000 owners band sits 10.0 above the median; the 0 .. 20,000 band trails by 5.00. In this extract, broader ownership tiers sit higher on Metascore relative to the file center than the sparsest shelf.

That gap is an association in the file, not proof that owners cause scores. It is still a citable structural pattern: critical mass and critical score lean together here.

Metascore and price form a joint market map

Metascore vs Price

Plotting Metascore against price shows clusters that averages erase — cheap high scores, expensive middling scores, and every bargain or premium pattern between.

The scatter is the right chart for the storefront question: what combinations of critic score and sticker price actually appear in the 26,688-row Steam Spy extract.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live Steam or Metacritic APIs. Missing Metascores, owner-band binning, price currency assumptions, and 2004–2018 coverage limits apply.

Findings describe this Steam Spy extract — structural signals about Metascore, owners, and price — not a complete quality ranking of every game ever shipped.

What to take away

Steam Spy scores in this file center on a median Metascore of 73.0, rise from 70.0 to 73.0 at the median, and peak at 98.0 among leaders such as Little Triangle.

Owner tiers matter: the 2–5 million band sits above the median while the smallest band trails, and Metascore–price clusters show how critic standing and sticker price co-occur on the shelf.

Sources

Data Science Learning Community. (2019). *TidyTuesday: Video Games Steam Spy*. https://raw.githubusercontent.com/rfordatascience/tidyuesday/main/data/2019/2019-07-30/video_games.csv

Editor's note

Artometrics data report from the TidyTuesday research pipeline. Charts and aggregates are reproducible from the embedded exhibits and public source files.

[View TidyTuesday source on GitHub](#)

Did Critical Acclaim Track Commercial Success in Video Games?

Steam-era concurrency is a living audience meter. This Sliced/TidyTuesday games extract holds 83,631 records spanning 2012–2021, with a median average player count (Avg) of 203 and a high of 1,584,887. PLAYERUNKNOWN'S BATTLEGROUNDS appears among the fact-box peaks; February is the most common month label in the file.

The charts ask how median Avg moved over time, which titles sit at the top, how months differ from the median, and whether Avg and Gain travel together. The calibration point is 203 — the typical observation in a market where a few live-service titles become cities unto themselves.

Fast facts

The numbers that set the scale for this report:

83,631Records in the working dataset
203Median Avg
1,584,887Highest observed Avg
PLAYERUNKNOWN'S BATTLEGROUNDTop Gamename by Avg
2012–2021Year span covered in the file
FebruaryMost common Month

Data and method

The source is the TidyTuesday release from 2021-03-16 (R for Data Science community). The working file contains 83,631 rows and 8 columns after merging available tables in the week folder. Avg is the primary metric; Gain appears in the scatter; month and game name are categorical axes.

Medians are mandatory because player counts are extremely skewed. Index-style fields are excluded from metric selection.

Median concurrency climbed across the decade

Median Avg Over Time

Median Avg rises from 212 in the opening period to 309 at the close — movement around and then above the file median of 203 depending on the window edges. The typical title-month in this extract ends higher than it began.

A rising median can reflect industry growth, survivor bias in which games remain observable, or both. The chart reports the center's path, not every title's fate.

Dota 2 leads a top tier that drops off fast

Dota 2 leads at 475,924 — 64,656 marks the median among the top dozen

Dota 2 leads at 475,924 Avg, while 64,656 marks the median among the top dozen. That plunge from first to the top-dozen median is the live-service economy in one sentence: a few games are continents; the rest of the elite tier is still large, but not the same species.

Fact boxes flag PLAYERUNKNOWN'S BATTLEGROUNDS among peak titles and 1,584,887 as the highest observed Avg — the extreme ceiling above even the charted leaders cut.

Months carry different concurrency distributions

Avg by Month

Category boxes by month show whether Avg consensus is shared or contested across the calendar. Seasonal launches, holidays, and school breaks can all reshape who is online.

February's frequency in the file is a coverage fact; the boxes ask a different question — how Avg spreads within each month.

January clears the median; October trails

Avg vs median by Month

January sits 36.9 above the median; October trails by 23.3. Those signed gaps turn the calendar into a comparable scale against the file center.

Month offsets are structural signals in this extract, not proof of a universal gaming season — but they are concrete enough to cite when asking when concurrency tends to run hot or cold.

Avg and Gain form a joint cloud of winners and churn

Avg vs Gain

Plotting Avg against Gain shows clusters that averages erase: high average with different gain profiles is a different audience story from low average with volatile gain.

The scatter is relational. It maps how level and change co-occur across the 83,631-row games slice.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live SteamCharts APIs. Missing values, title variants, and 2012–2021 coverage limits apply. Merged tables may fan out or duplicate rows when join keys are imperfect.

Avg and Gain are the metrics as defined in the source. Findings describe this video-games extract — concurrency structure — not a complete commercial P&L for every title.

What to take away

Concurrency in this file centers on a median Avg of 203, rises from 212 to 309 at the median, and still supports extremes above a million players.

Dota 2 leads the charted ladder, January sits above the median while October trails, and Avg-Gain clusters show how level and change travel together in a long-tailed live-service market.

Sources

Data Science Learning Community. (2021). *TidyTuesday: Video Games*. <https://raw.githubusercontent.com/rfordatascience/tidyuesday/main/data/2021/2021-03-16/sliced-tidyuesday.csv>

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What Pokémon's Franchise Actually Comprises

Pokémon is the highest-grossing media franchise in history — more than \$150 billion in lifetime revenue across games, cards, merchandise, and film. That number gets cited constantly. What gets cited less is the design infrastructure underneath it: the 18-type system, the base stat framework, the generation-by-generation release cadence that has added roughly 100 new Pokémon every few years since 1996. The franchise looks like a cultural phenomenon. The data shows it's also an engineering project.

The TidyTuesday 2025-04-01 Pokémon dataset — sourced from PokéAPI and covering every Pokémon through Generation IX — lets three questions land cleanly. Do different types actually play differ-

ently, or is that a perception? Has Game Freak been inflating Pokémon power over time? And what does physical size reveal about how Pokémon are designed?

Research question

Can Pokémon's creature design be measured as a system rather than described only as franchise lore? This report asks whether primary type, generation, base experience, height, and weight reveal consistent design rules across the PokéAPI-derived TidyTuesday dataset.

The observational question is practical: do Dragon, Steel, Fighting, Bug, Water, and other types carry statistically distinct stat identities; does the upper tail of base experience creep across generations; and do physical dimensions follow a coherent size grammar with deliberate exceptions such as Cosmoem and Wailord?

Fast facts

The numbers that set the scale for this report:

949 Pokémon in the dataset with a confirmed primary type
18 Distinct primary types represented
8 Generation IDs in the data — Gen 8 and 9 unclassified in source
157 Median base experience across all Pokémon
510 Pokémon with a second type — the dual-type majority
Water Most common primary type in the dataset

Data and method

The dataset is TidyTuesday Week 13, 2025 — a cleaned version of the PokéAPI database covering 1,028 Pokémon entries across nine generations. Each row represents a single Pokémon form, which means alternate forms (Mega Evolutions, regional variants, Gigantamax forms) appear as separate records from their base versions. The core fields used in this piece are the six base stats (HP, Attack, Defense, Special Attack, Special Defense, Speed), base experience, primary type, height, weight, and generation ID. All values are drawn directly from the TidyTuesday source — no external APIs were joined for this analysis.

Base experience is a commonly misunderstood field. It is not the same as a Pokémon's base stat total, and it is not a measure of overall power. It is the amount of experience a trainer's Pokémon receives when defeating that species in battle — a game mechanic that reflects intended difficulty and progression pacing, not raw strength. A Pokémon with high base experience is meant to be encountered later in the game or to signal a harder fight. It correlates with power, but imperfectly.

Height and weight are stored in the dataset in their PokéAPI native units: decimetres for height and hectograms for weight. Both were converted to standard metric units (metres and kilograms respectively) before plotting. The log scale applied in Chart 3 is not a stylistic choice — the raw distributions span roughly four orders of magnitude, from Flabébé at 0.1 metres and 0.1 kilograms to Wailord at 14.5 metres and Cosmoem at 999.9 kilograms. A linear scale would make the chart unreadable.

Stat Identity By Type

Stat Heatmap

Dragon sits at the top of the heatmap for a reason. With a median Attack of 100 and no single stat below 80, Dragon-types are the closest thing Pokémon has to a generalist elite — high across the board, weak nowhere. The type wasn't always rare by accident. Game Freak made Dragon deliberately difficult to access in the early games because the numbers justified the gatekeeping.

The more revealing story is in the specialists. Steel-types clock a median Defense of 116 — the highest single cell on the entire chart — while their Special Attack sits at 60, dead last among Steel's own stats. Fighting-types mirror that tradeoff on offense: Attack of 100, Special Attack of 42, the sharpest stat gap in the dataset. These aren't balanced Pokémon. They're designed to do one thing extremely well and pay for it everywhere else. The heatmap makes that design philosophy impossible to ignore.

Bug sits alone at the bottom. Across all six stats, Bug-types post the lowest or near-lowest medians in every column — 60 HP, 65 Attack, 60 Defense, 55 Special Attack, 60 Special Defense, 60 Speed. The type has been statistically undertuned since Generation I and the data across nine generations confirms it hasn't been corrected. Whether that's a deliberate choice to keep early-game Pokémon weak or an oversight

that calcified into canon, the result is the same: Bug-types are the worst-statted primary type in the game by a significant margin. One caveat worth noting: Flying appears near the top of the chart with an unusually high Speed median of 116, but Flying is almost never a primary type — it's almost always secondary. The dataset's primary Flying entries (Tornadus, Gyarados, Chatot, and a handful of others) are a small and unrepresentative sample, which inflates the type's apparent standing relative to how Flying Pokémon actually play in practice.

Generation Creep

Generation Ridgeline

The ridgeline doesn't lie. Every generation from Gen 1 through Gen 7 shows the same basic distribution shape — a peak around 60–70 base experience on the left, a second hump around 150–170 for mid-tier Pokémon, and a long tail trailing off toward the high end. What changes is that tail. With each successive generation, it stretches a little further right. The center of mass isn't dramatically shifting, but the ceiling keeps rising.

The outliers tell the real story. That thin red thread extending past 600 on Gen 2's row is Blissey — a Normal-type with 255 base HP and the highest base experience yield of any Pokémon in the game at the time of its introduction. Gen 1 has its own isolated tail around 400, anchored by Chansey and the original legendary trio. These aren't flukes of the distribution — they're intentional design decisions that show up as statistical anomalies. Game Freak wasn't inflating power evenly across all Pokémon. They were adding extreme outliers at the top end while keeping the median relatively stable.

Gen 7's distribution is the most compressed of any generation — the narrowest spread, the sharpest peak, the least variance. That's partly a function of Game Freak introducing fewer fully-evolved Pokémon with extreme stat totals in Alola, and partly a reflection of a design era that prioritized Mega Evolutions and Z-Moves over raw base stat inflation. The shape of Gen 7's curve is what controlled power creep looks like. Whether Game Freak intended it or stumbled into it, the ridgeline is the evidence.

Built Different

Size Scatter

The first thing the scatter reveals is that Pokémon size follows a rule: heavier things are taller, and the relationship is consistent enough across 1,000+ Pokémon that the log-scale correlation is basically a straight line. Game Freak didn't design each Pokémon's physical dimensions independently — there's an implicit size grammar baked into the franchise, and almost every Pokémon obeys it. The exceptions are the entire point of the chart.

Cosmoem is the most extreme outlier in the dataset — 999.9 kg crammed into a body 0.1 meters tall, which puts it in the bottom-right corner of the chart completely alone. It weighs more than a polar bear and is smaller than a dinner plate. Wailord goes the other direction: 14.5 meters tall, 398 kg, making it one of the least dense objects in the Pokémon world despite being a whale. Celesteela, a Steel/Flying Ultra Beast, is nearly as heavy as Wailord and taller than a four-story building. These aren't design oversights — they're deliberate choices to make certain Pokémon physically impossible in ways that signal their otherness before you ever check their stats.

The type coloring doesn't produce clean clusters, but it does reveal soft patterns. Dragon-types (purple) and Water-types (blue) dominate the upper-right of the chart — the large, heavy end of the distribution. Bug-types (yellow-green) cluster densely in the small, light region on the left. Steel-types scatter across an unusually wide range, from the tiny Klefki to the massive Celesteela. What the scatter ultimately shows is that physical scale is a storytelling tool. The Pokémon at the extremes aren't there because of random number generation — they're there because someone at Game Freak decided that a 14-meter whale or a 999-kilogram cosmic cocoon needed to exist, and the data is just the record of that decision.

What this file cannot tell you

The most significant limitation of this analysis is scope: the dataset covers base stats and physical attributes, but Pokémon competitive viability is determined by factors this data doesn't capture. Abilities, move pools, type matchups, held items, and EV/IV systems all affect how a Pokémon actually performs in battle. A Bug-type with low base

stats can still be competitively viable if it has the right ability — Scizor has been a top-tier competitive Pokémon for years despite its type disadvantage, largely due to its access to Bullet Punch and Technician. The numbers here describe design intent, not gameplay outcomes.

The generation variable has a known data quality issue: a subset of records carry null generation IDs, most likely alternate forms or entries added outside the standard generational release structure. These were filtered from Chart 2 rather than imputed, which means the ridgeline distributions reflect only records with confirmed generation assignments. The omitted records are a small fraction of the total dataset and are unlikely to meaningfully change the shape of any generation's distribution.

Finally, the dataset reflects a single snapshot of PokéAPI data compiled for the TidyTuesday release in April 2025. Base stats, forms, and classifications are subject to change with new game releases and balance patches — particularly for Scarlet and Violet, which introduced a large number of new forms and mechanics that may not be fully represented in this version of the data. Any analysis of the most recent generations should be treated as preliminary.

What to take away

The type system isn't just flavor. Dragon-types are statistically elite by design, Bug-types are statistically weak by design, and the specialists in between — Steel's defense, Fighting's attack, Electric's speed — reflect intentional tradeoffs embedded in the numbers from the beginning. The heatmap makes 25 years of Game Freak's design philosophy legible in a single chart. Every type has a signature, and the data confirms it.

Power creep is real but it's not uniform. The median Pokémon hasn't gotten dramatically more powerful across nine generations — the distribution shape is remarkably stable from Gen 1 to Gen 7. What has changed is the ceiling. Each generation adds a small number of outliers at the extreme high end: a Blissey, a legendary trio, an Ultra Beast. The franchise inflates not by pulling the average up but by making the top more extreme. That's a different kind of creep, and it's harder to notice until you plot it.

The size scatter is the most human chart of the three. It shows that somewhere inside Game Freak, someone made a decision to create a whale that weighs less than a car and a cosmic dust mote that weighs

more than a horse. Those aren't statistical accidents — they're creative decisions that left a data signature. That's the throughline of this entire report: the Pokémon franchise is enormous and commercially optimized and thoroughly analyzed by millions of fans, but underneath all of it is a set of design choices made by people, and the numbers are how you read them.

Sources

The PokéAPI/TidyTuesday file is useful because it exposes the game-design schema behind the franchise: types, stats, generation IDs, dimensions, and battle-related mechanics in a machine-readable form. It is less useful for competitive outcomes because abilities, movesets, items, effort values, individual values, metagame bans, and format-specific rules sit outside the columns plotted here. The report therefore treats the data as design infrastructure, not as a live Smogon, VGC, or Pokémon Showdown viability model.

The historical anchors are Satoshi Tajiri, Game Freak, Nintendo, Creatures, and The Pokémon Company because the numbers emerge from a designed commercial system that began with *Pocket Monsters Red and Green* in 1996. Bulbapedia and PokéAPI provide community and API documentation for mechanics such as base experience, but the interpretation remains tied to the official games' release cadence and the dataset snapshot rather than to fan speculation about future balance changes.

Data Science Learning Community. (2025). *TidyTuesday: A weekly social data project*. Week 13, 2025 — Pokémon. <https://tidytues.day>

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PokéAPI. (2025). *The RESTful Pokémon API*. <https://pokeapi.co>

Bulbapedia contributors. (2025). *Base experience*. Bulbapedia, the community-driven Pokémon encyclopedia. https://bulbapedia.bulbagarden.net/wiki/Base_experience

Editor's note

This report was researched, written, designed, and produced in active collaboration with Claude AI (Anthropic). The data pipeline, statistical analysis, chart design, written analysis, narrative structure, and visual styling were all developed through a directed partnership between human editorial judgment and AI execution.

Artometrics was built on the premise that rigorous analysis and honest process are not in conflict. The research questions, editorial instincts, interpretive framing, and brand vision are ours. The execution — every line of R code, every paragraph of analysis, every design decision — was a collaboration. We document this not as a disclaimer but as a description of how we actually work, and as a position: we believe this is what serious data journalism looks like when the tools available are used honestly and at full capacity.

— *Artometrics Editorial*

[View source on GitHub](#)

Which Board Games Combine Crowd Scale With Elite Ratings?

BoardGameGeek ratings are a public taste machine: tens of thousands of titles, each scored by the people willing to log plays. The TidyTuesday board-games extract used here holds **10,532** records with a median average rating of **6.39** and a high of **9.00** for Small World Designer Edition.

Wargame, World War II is the most common category label in the file. That already tilts the catalog toward conflict simulations — a hobby subculture with its own rating norms. The charts ask how ratings split by category, how leaders separate from the median, and whether crowdfunding-scale user counts track with elite scores.

Fast facts

The numbers that set the scale for this report:

10,532Records in the working dataset
6.39Median Average rating
9.00Highest observed Average rating
Small World Designer EditionTop Name by Average rating
1970–1970Year span covered in the file
Wargame,World War IIMost common Category

Data and method

The source is the TidyTuesday release from 2019-03-12 (board_games.csv). After cleaning, 10,532 rows remain in the working frame.

Average rating is the primary metric. Category fields can be multi-label strings. Charts are Plotly JSON with PNG fallbacks. The year span encoded in the fact box reflects the filtered time field present in this build of the file.

The Catalog's Median Rating Sits Near 6.39

Median Average rating Over Time

The time view in this extract collapses to a median average rating of about **6.39** — the same calibration point as the fact box. In BoardGameGeek terms, that mid-6s band is ordinary: playable, not legendary.

Everything that follows should be read against that baseline. A 9.00 specialty edition is not “a bit better than average”; it is a different prestige tier.

Designer Editions and Heavy Games Own the Peak

Small World Designer Edition leads at 9.00 — 8.78 marks the median among the top dozen

Small World Designer Edition leads at **9.00**. Kingdom Death: Monster (~**8.93**), Terra Mystica: Big Box (~**8.85**), and a run of deep wargames — Last Chance for Victory, The Greatest Day, Last Blitzkrieg, Enemy Action: Ardennes — fill out the top dozen. The median among those leaders is about **8.78**.

These are not mass-market party games. They are hobby-complete objects — big boxes, campaign systems, hex-and-counter epics — rated by audiences selected for intensity.

Wargame Categories Rate Higher Than Card Games

Average rating by Category

Category boxes show Napoleonic wargames with a median near **7.04** (n=124) and World War II wargames near **6.89** (n=449). Card games sit lower, with a median near **6.03** (n=438). Abstract strategy lands near **6.28** (n=284).

Volume and prestige diverge: card games are numerous but center closer to the catalog median; dedicated wargame labels clear a higher bar among the people who rate them.

Civil War and Napoleonic Titles Beat the Median Most

Average rating vs median by Category

Gaps to the overall median are largest for American Civil War wargames (~**+0.66**) and Napoleonic wargames (~**+0.65**), with World War II wargames also positive. Dice and generic card-game labels sit below the median.

The pattern is consistent: historical conflict simulations, as tagged in this file, are the categories that most reliably clear the 6.39 baseline.

Crowd Size and Rating Are Related — Loosely

Average rating vs Users rated

Average rating versus users rated shows popular hits with tens of thousands of ratings landing across the mid-to-high 7s, while many high-rated titles still have modest user counts. Elite scores without crowd scale remain possible.

The scatter’s lesson is citable: popularity predicts attention, not a precise quality score. A 9.00 designer edition and a 7.4 smash hit solve different problems in the hobby economy.

What this file cannot tell you

BoardGameGeek ratings are self-selected by hobbyists; children’s games and mass-market titles can be systematically under- or over-represented. Multi-label categories double-count games across boxes.

Expansions and designer editions inflate prestige relative to base games. The file is a 2019 community snapshot, not a live BGG API pull.

What to take away

Across 10,532 games, the median rating is 6.39; the top dozen lives near 8.8–9.0. Wargame categories clear the baseline more often than broad card-game labels.

Cite 6.39 as the catalog’s center of gravity, and treat 9.00 specialty editions as a separate prestige economy — not as proof that “games got better.”

Sources

Data Science Learning Community. (2019). *TidyTuesday: Board Games*. https://raw.githubusercontent.com/rfordatascience/tidyuesday/main/data/2019/2019-03-12/board_games.csv

What a Century of Anime Titles Reveals About the Industry

This MyAnimeList dataset covers 13,631 unique anime titles spanning 1917 to 2019 — a century-scale archive tracing the medium from pre-war theater shorts to the standardized streaming machine of the 2010s. The key interpretive move here is separating catalog reality (what actually got made) from fan-canon popularity (what people actually watched and rated). Those two things rhyme, but they are not the same thing, and the gap between them is where the most interesting analysis lives.

The overall median MAL score is 6.38. That is the calibration point for everything that follows. A 6.38 is not a failure — it is the mathematical center of a platform where over 13,000 titles compete for attention. When comparing studios, formats, and genres throughout this piece, the question is never who broke 8.0 once. It is who stayed above 6.38 consistently, across volume. That is a far harder problem, and most studios never solve it.

Research question

What does a century-scale MyAnimeList catalog reveal about anime as an industrial system rather than a fan canon? This report asks how release formats, studio consistency, genre popularity, and user ratings separate what the industry produced from what platform users later rewarded.

The observational question is whether distribution channels and production institutions leave a measurable signature in the catalog: TV versus OVA versus ONA, high-volume studios versus boutique studios, and genres that are popular without being highly rated or highly rated without becoming algorithmically visible.

Fast facts

The numbers that set the scale for this report:

13,631 Unique anime titles in the dataset — from silent-era shorts to modern streaming originals

6.38 Overall median MAL score — the mathematical baseline every studio and genre is measured against
8.12% Share of TV anime scoring 8+ on MAL — the highest prestige hit-rate of any format (346 out of 4,260 titles)
90 High-scoring shows buried past popularity rank 2,000 — the titles the algorithm never surfaced
0.389 Pearson correlation between score and members — popularity predicts quality directionally, not precisely
2006 The Haruhi inflection point — when light novels became the anime industry's primary outsourced R&D pipeline

Data and method

The source data is the MyAnimeList TidyTuesday release from April 23, 2019, maintained by the R for Data Science community. It contains scraped records for 13,631 unique anime titles with fields for format type, source material, episode count, content rating, genre tags, studio attribution, MAL score, member count, favorites count, and air date range. The data covers titles from 1917 through early 2019.

The dataset requires meaningful cleaning before analysis. Air dates arrive as unstructured strings and must be parsed via regex to extract start and end year. Genre, studio, and producer fields are stored as Python-style list strings — bracket-delimited, comma-separated — that must be split and unnested into long-format tables for aggregation. Any title with a rank or popularity of zero is excluded, as these represent incomplete records without enough community engagement to produce reliable metrics.

Scores on MAL are user-submitted on a 1–10 scale. The platform has a known self-selection bias: its userbase skews adult, male, and English-speaking relative to anime's actual global audience. This produces systematic underscoring of children's content and promotional music videos, and potential overrepresentation of properties with strong Western fandom overlap. The dataset also predates the post-2019 streaming era — *Demon Slayer*, *Jujutsu Kaisen*, *Chainsaw Man* are absent — which means any claims about current genre trends require updating.

An Industry That Industrialized — Then Diversified

Releases By Year

The release history of anime is not a single growth story — it is six separate ones running in parallel. Movies were the dominant format from the 1920s through the 1970s, a low-volume craft economy built for theater. The first structural break arrived in the late 1980s, when the OVA boom gave creators a direct-to-consumer channel that bypassed TV censors entirely. OVA peaked around 1990, then collapsed as TV economics took over.

TV is the modern engine. It industrialized in the late 1990s and has compounded steadily since. Specials peaked in the mid-2000s as a franchise extension format, then plateaued. Music videos spiked dramatically in the 2010s — largely driven by short promotional content — and then fell just as fast. The most important line on this chart is ONA : near-zero before 2010, then a vertical ascent. That is not a genre trend. It is a platform shift. Netflix, Crunchyroll, and YouTube did not just distribute anime — they created a new format category.

What the faceted view makes clear is that these formats are not competing for the same audience at the same time. They are sequential industrial experiments. Each one dominated its era, then either stabilized into a niche or was superseded by the next model. The industry did not simply grow — it restructured itself around whichever distribution channel controlled access to viewers.

Bones and Kyoani Lead — Consistency Across Volume Is the Real Achievement

Studio Consistency

Every studio on this chart has produced at least one great show. That is not the achievement. The achievement is the IQR bar — the horizontal line showing the spread between the 25th and 75th percentile of a studio's scores. A tight bar at a high median means the studio's worst output is nearly as good as its best. Bones (n=115) leads in median and maintains a tight IQR across a massive catalog. White Fox (n=33) has

the highest median dot but a wide IQR — the boutique model: fewer swings, but when they miss, they miss. Kyoto Animation (n=110) is the most consistent large studio on the chart: 110 titles with a tight IQR anchored well above the 6.38 baseline.

Trigger (n=23) has the widest IQR — spanning from near the baseline to above 8.0. High-risk, high-reward. Studio Deen (n=264) is the industrial cautionary tale: the largest catalog on the chart and the lowest median. Volume without curation dilutes quality. A-1 Pictures (n=190) sits in the middle — a production machine that delivers reliable output at scale, occasionally punctuated by genuine hits.

The viridis color scale maps TV share. Studios with higher TV share cluster in the middle — TV volume is the primary dilution mechanism. Boutique studios with low TV share show more extreme medians in both directions. The 6.38 dashed line is the dataset median. Every studio above it is beating the field. Most of them are doing it with fewer than 50 titles — which tells you exactly how hard it is to stay above average at scale.

Genre Map: What's Popular vs What's Well-rated?

Genre Map

The genre map plots every major genre against two axes: how popular its titles are (X, log-reversed so most popular is right), and how well-rated they are (Y). Size encodes volume. The Main Hall — Action, Comedy, Fantasy, Adventure — sits in the dense center-right cluster. Massive reach, enormous volume, medians diluted to the mid-6s by sheer output. These are the genres that define what most people think anime is.

The Curator's Rooms are the upper-left quadrant. Thriller sits alone at the top: highest median on the chart (~7.5), extremely low popularity rank. The genre barely exists by volume, but almost everything made in it is exceptional. Mystery and Psychological follow the same pattern — high score, mid-obscurity — which aligns with their SQL Query 2 performance. Historical sits far right: highly rated but ignored algorithmically despite strong critical reception. The starkest bottom outlier is Dementia : low popularity, low score, tiny volume.

Kids and Music are the bottom-right anchors — widest audiences, lowest scores. This is the self-selection problem in reverse: the MAL platform's adult userbase applying adult critical frameworks to children's media and promotional content. The scores are not wrong given who is rating them. They are simply the wrong instrument for the material. The genre map's quadrant structure is the clearest single visualization of what the platform is: a tool built by and for a specific audience, measuring everything — including content that was never made for them — against that audience's preferences.

What this file cannot tell you

The dataset ends in April 2019. *Demon Slayer*, *Jujutsu Kaisen*, *Chain-saw Man*, *Spy × Family* are entirely absent. Their exclusion almost certainly understates ONA growth, understates action/Shounen dominance, and means any claims about current genre trends require updating against the post-2019 streaming landscape. The industry restructured significantly around Netflix and Crunchyroll originals after this cutoff — the format and source material charts would look different with five more years of data.

MyAnimeList is an enthusiast platform. Its userbase skews adult, male, and English-speaking relative to anime's actual global audience. Two direct consequences appear in the data: Kids and Music programming are systematically underscored because the rating population is not the target demographic, and shows with strong Western fandom overlap (*Fullmetal Alchemist: Brotherhood*, *Death Note*) may be overrepresented in the elite tiers. The platform measures what its users value, which is not the same as what anime's full global audience values.

Genre tags are not mutually exclusive, and studio medians for boutique operations ($n < 30$) are statistically fragile. White Fox's strong studio ranking rests on 33 titles — a single poor season could meaningfully shift it. Thriller's top-ranked genre position rests on a small title count. Both results are directionally interesting and statistically fragile, and should be read as signals worth investigating rather than settled conclusions.

What to take away

Anime transformed its distribution model three times in a century: from a theatrical craft economy built for prewar audiences, to a home-video OVA market that bypassed broadcast entirely, to an industrial-

ized TV machine running on seasonal 12-episode contracts, and finally into a diversified platform ecosystem where ONA and streaming have created a fourth distribution channel that did not exist before 2010. Each transition was driven by access — whoever controlled the path to viewers controlled the format that dominated production.

Through all of it, two structural facts held. TV is the primary engine for quality output — 8.12% hit-rate, the highest of any format, confirmed by both the SQL aggregation and the genre map's quadrant structure. And popularity and quality are correlated but not causal — the feedback loop runs both directions, the algorithm amplifies existing popularity rather than discovering hidden quality, and the score-members correlation of $r = 0.389$ confirms the relationship is real but far from deterministic.

Ninety high-scoring titles sit past popularity rank 2,000 — past the point any algorithm will surface them. The catalog is vast. The attention economy is ruthless. If you want the best statistical odds, shop the seasonal TV hits in Mystery, Shounen, or Romance — the three genres with the highest average scores above 500 titles. But if you are willing to walk past the algorithm's edge, there are 90 titles waiting with scores above 8.0 and audiences small enough that they never quite made the trending list.

Sources

The MyAnimeList release should be read as a platform archive, not a census of Japanese animation. Its fields preserve what MAL users and scrapers could see by April 2019: title metadata, format labels, source material, score, members, favorites, and genre tags. That makes it unusually useful for studying fan attention and catalog structure, while also making it incomplete for post-2019 streaming shifts, Japanese domestic audience behavior, and theatrical or children's programming underrepresented on English-language enthusiast platforms.

The interpretive frame draws on media-industry work by Ian Condry, Rayna Denison, Marc Steinberg, and Jonathan Clements because the dataset's statistical patterns map onto known institutional changes: television production committees, light-novel and manga source pipelines, global streaming distribution, and the media-mix system that turns characters and franchises into cross-platform assets. Those citations help keep the charts grounded in actual anime-industry history rather than treating MAL scores as free-floating taste metrics.

MyAnimeList Dataset (2019). TidyTuesday, April 23, 2019. Retrieved from: <https://github.com/rfordatascience/tidytuesday/tree/main/data/2019/2019-04-23>

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Editor's note

This report was researched, written, designed, and produced in active collaboration with Claude AI (Anthropic). The data pipeline, statistical analysis, chart design, written analysis, narrative structure, and visual styling were all developed through a directed partnership between human editorial judgment and AI execution. The research questions and brand vision are ours. The execution is a collaboration.

— Artometrics Editorial

[View source on GitHub](#)

How LEGO Set Complexity Evolved Across Themes

LEGO sets are toys, collectibles, and increasingly architectural objects. The Rebrickable-style database that feeds this TidyTuesday release turns that catalog into something measurable: parts counts, set names, and how complexity has shifted across decades.

The working file's fact markers put the median set near **34** parts while the extreme end reaches into five digits — **World Map** leads at **11,695** parts on the highlighted ranking, with **5,792** as the median among the top dozen. Across the time span in the charts, median part counts fall from roughly **142** in the opening period to about **59.5** at the close — a reminder that the typical set and the showcase set are different products.

Quantity of pieces is not the same as play value, but it is a clean industrial signal: how much plastic inventory a box commits to, and how the company balances everyday sets against display pieces.

Fast facts

The numbers that set the scale for this report:

19,798Records in the working dataset
34.0Median Num parts
11,695Highest observed Num parts
World MapTop Name by Num parts
1949–2022Year span covered in the file

Data and method

The source is the TidyTuesday release from 2022-09-06 (R for Data Science community). The working file contains 19,798 rows across set and parts tables after merging — set names, years, part counts, and quantity fields among them.

Medians are essential because a few architectural and mosaic sets pull means upward. Charts export as Plotly JSON with PNG fallbacks. Part counts describe inventory complexity, not retail price or build time directly.

How part counts changed over time

Median Num parts Over Time

Median part counts fall from about **142** in the opening period of the series to about **59.5** at the close. That does not mean LEGO abandoned complexity — it means the typical catalog entry became smaller even as a new class of ultra-large display sets appeared at the extreme.

A falling median beside a rising ceiling is a classic catalog split: more small sets for everyday buyers, and a thinner tier of destination builds for adults and collectors.

Who sits at the top

World Map leads at 11,695 — 5,792 marks the median among the top dozen

World Map leads at **11,695** parts. The median among the top dozen is **5,792** — still orders of magnitude above the file-wide median near 34. Names such as NINJAGO City, Hogwarts Express Collectors' Edition, stadium sets, Taj Mahal, Colosseum, and Titanic populate the same high-complexity band in related leader cuts.

These are not children's starter boxes. They are furniture-scale plastic architecture sold under a toy brand — and the part counts make that product shift quantitative.

How the field is spread

Num parts Distribution

The distribution is sharply right-skewed: median about **34** versus mean about **161**. The top decile begins near **432** parts. Most sets are modest; a long right tail carries the Instagram-era display pieces.

Anyone quoting the mean as the typical LEGO set is describing a more complex object than most boxes on the shelf. The median is the honest center; the tail is the brand's adult collector strategy made visible.

Leader trends

Top Name Over Time

The leading names do not move in lockstep. Some franchise icons fade as others surge; licensed themes and modular city sets take turns occupying the complexity frontier.

Tracking leaders over time separates sustained flagship lines from one-off mosaic promotions. Both matter commercially; only the sustained lines reshape what LEGO means as a cultural object year after year.

Parts and quantity

Num parts vs Quantity

Plotting number of parts against quantity fields shows clusters that averages erase. Some rows describe unique large sets; others describe common small parts used in volume across many builds.

The scatter is a reminder that the database mixes set-level and parts-level grains depending on the joined table. Complexity at the box level and repetition at the brick level are related industrial facts, not the same metric.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and release coverage limits apply. Part counts are not prices, and set names can recur across regions or reissues.

Findings describe structural signals about LEGO catalog complexity — not a full financial history of the company or a ranking of which sets are best to build.

What to take away

LEGO's catalog is two products sharing one logo: a dense middle of small sets around a few dozen parts, and a thin upper tier of multi-thousand-piece display builds that redefine the brand's adult market.

The citable tension is the falling median beside the extreme ceiling. Everyday LEGO got smaller in the typical case even as the largest boxes became architectural.

Sources

Who Ranks Highest on Comic-Book Power Scales?

Comic-book universes are attention economies with capes. The TidyTuesday comic-characters extract used here holds **23,272** records spanning **1935–2013**, with a median of **4.00** appearances and a high of **4,043** for Spider-Man (Peter Parker). Marvel is the most common publisher label.

That median of four appearances is the catalog's quiet truth: most named characters are spear-carriers. The charts follow how appearance mass concentrates in a handful of franchise engines.

Fast facts

The numbers that set the scale for this report:

23,272Records in the working dataset

4.00Median Appearances

4,043Highest observed Appearances

Spider-Man (Peter Parker)Top Name by Appearances

1935–2013Year span covered in the file

MarvelMost common Publisher

Data and method

The source is the TidyTuesday release from 2018-05-29 (week9_comic_characters.csv). After cleaning, 23,272 rows remain.

Appearances is the primary metric; publisher splits Marvel versus DC; Pareto charts show concentration among leading names. Charts are Plotly JSON with PNG fallbacks.

Early Characters Logged Higher Median Appearances

Median Appearances Over Time

Median appearances by first-appearance year peak in the mid-1930s (about **125** in 1935, **117** in 1937) and fall toward single digits for many later decades as the catalog explodes with short-lived names.

Golden Age cohorts look “busier” partly because survivors were the ones who kept being reprinted and reintroduced. Later decades add more characters who never become franchise centers.

A Dozen Icons Own the Appearance Summit

Spider-Man (Peter Parker) leads at 4,043 — 2,377 marks the median among the top dozen

Spider-Man leads at **4,043** appearances, followed by Captain America (**3,360**), Batman (**3,093**), Wolverine (**3,061**), Iron Man (**2,961**), and Superman (**2,496**). Thor and Ben Grimm continue the elite band. The median among the top dozen is about **2,377**.

These are not merely popular characters; they are publishing infrastructure — identities that can carry ongoing series, crossovers, and adaptations.

Marvel Has More Characters; DC's Median Appearances Run Higher

Appearances by Publisher

Marvel contributes about **15,280** boxed rows with a median of **3** appearances and a max of 4,043. DC contributes about **6,541** rows with a higher median of **6** and a max of 3,093 (Batman).

Marvel's catalog is wider; DC's typical named character in this file has been used more often. Volume and intensity are different publisher strategies.

The Top Five Names Hold Nearly Half the Leader Aggregate

The top 5 name entries account for 45% of the aggregate appearances

On the Pareto curve of leading names, the top five entries account for about **45%** of the aggregate appearances in that ranked view. Share accumulates steeply from there toward the fifteenth name.

Franchise comics are a power-law business. A short list of identities absorbs a huge fraction of cumulative attention.

Concentration Repeats as a Structural Fact

The top 5 name entries account for 45% of the aggregate appearances

The second concentration chart restates the same Pareto shape: top five near 45%, with the curve climbing toward 100% by the fifteenth plotted name.

Repeating the concentration view is the point — this is not a fragile binning artifact. Appearance mass in superhero publishing is systematically top-heavy.

What this file cannot tell you

Appearance counts depend on how databases credit cameos, alternate Earths, and reboots. Publisher field collapses imprints. The 2013 end-point misses the full MCU/DCEU feedback loop on print appearances.

Median appearances of 4 should not be misread as “characters only show up four times then vanish” without checking how incomplete rows are handled.

What to take away

Across 23,272 characters, the median appearance count is 4, while Spider-Man exceeds 4,000. The top five leaders hold about 45% of the ranked appearance aggregate.

Cite Marvel’s wider catalog versus DC’s higher median when comparing publishers, and treat the dozen icons above ~2,300 appearances as the industry’s real capital stock.

Sources

Data Science Learning Community. (2018). *TidyTuesday: Comic Book Characters*. https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2018/2018-05-29/week9_comic_characters.csv